DE LA SALLE UNIVERSITY – DASMARIÑAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS
Second Semester, SY 2024-2025

COURSE CODE	**:**	S-ITNT323
COURSE TITLE	**:**	Cisco Networking Semester 2 - SRWE
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	Two Units
LEARNING MODALITY	**:**	Traditional Blended Learning
PRE-REQUISITES	**:**	S-ITNT312 Lec and Lab
CO-REQUISITES	**:**	S-ITNT323Lab
PROFESSOR	**:**	Maria Gloria Del Rosario
CONSULTATION HOURS	**:**	Tuesday 11:00 – 5:00 online Friday: 1:00-4:00 COS100C
CONTACT INFORMATON	**:**	madelrosario@dlsud.edu.ph (046) 4811900 local 3134

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course focuses on the fundamental networking concepts and technologies. Cisco Certified Networking Associate (CCNA) course incorporates theoretical aspects and practical application while providing students the skills and hands-on experience needed to design, install, operate and maintain networks. The course covers a complete range of networking concepts, and features many hands-on projects where students have access to the building blocks of Local and Wide Area Networks (LANs and WANs), such as wireless access points, switches and routers.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. Configure VLANs and Inter-VLAN routing applying security best practices.
CLO2. Troubleshoot inter-VLAN routing on Layer 3 devices.
CLO3. Configure redundancy on a switched network using STP and EtherChannel.
CLO4. Troubleshoot EtherChannel on switched networks.
CLO5. Explain how to support available and reliable networks using dynamic addressing and first-hop redundancy protocols.
CLO6. Configure dynamic address allocation in IPv6 networks.
CLO7. Configure WLANs using a WLC and L2 security best practices.
CLO8. Configure switch security to mitigate LAN attacks.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Configure VLANs and Inter-VLAN routing applying security best practices.	TLO1. Configure initial settings on a Cisco switch
CLO2. Troubleshoot inter-VLAN routing on Layer 3 devices.	TLO2. Explain how frames are forwarded in a switched network.
CLO3. Configure redundancy on a switched network using STP and EtherChannel.	TLO3. Explain how STP operates in a simple, switched network.
CLO4. Troubleshoot EtherChannel on switched networks.	TLO4. Describe Ethernet technology.

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Onsite / F2F	Asynchronous	Allotted Hours
Module 0 CLO1,2 TLO 1,2	Week 1 Jan 20-25	Gospel Reading, Reflection, and Prayer Discussion of Course Syllabus Presentation on the Traditional blended learning model LMS: 1. Schoolbook enlistment 2. Netacad.com enrolment 3. MS Teams channel for alternative communication mode Module 0 – Basic Device Configuration Interface Verification Commands Verify Interface Status Verify Directly Connected Networks Download Packet Tracer version 8.2.2	netacad.com MS Teams SchoolBook	ü		2
Module 1 CLO1,2 TLO 1,2	Week 2 Jan 27-Feb 1	Gospel Reading, Reflection, and Prayer Module 1 Routing Protocols Compare Static and Dynamic Routing Concepts Enabling Activity 1 – Quiz on Routing Concepts	netacad.com MS Teams SchoolBook	ü		2
Module 2 CLO1,2 TLO 1,2	Week 3 Feb 3 - 8	Gospel Reading, Reflection, and Prayer Module 2 IP Static Routing Routing Table Path Determination Enabling Activity 2 – Routing Concepts and Configuration Exam	netacad.com MS Teams SchoolBook	ü		2
Module 2 CLO1,2 TLO 1,2	Week 4 Feb 10-15	Gospel Reading, Reflection, and Prayer Module 2 – Troubleshoot Static and Default Routes Configure and Re-configure routing protocols Class Participation1 – Checkpoint Exam	netacad.com MS Teams SchoolBook	ü		2
Module 3 CLO 1,2,3 TLO 2,3,4	Week 5 Feb 17-22	Gospel Reading, Reflection, and Prayer Module 3 – VLANs Explain the purpose of VLAN in a switched enviroment Enabling Activity 3 – Configure VLANs	netacad.com MS Teams SchoolBook		ü	2

Module 3 CLO 1,2,3 TLO 2,3,4	Week 6 Feb 24-Mar 1	Gospel Reading, Reflection, and Prayer Module 3 – Inter-VLAN Routing Describe the options for configuring Inter-VLANs routing Class Participation2 – Checkpoint Exam	netacad.com MS Teams SchoolBook		ü	2
Module 4 CLO 3,4 TLO 3,4	Week 7 Mar 3-8	Gospel Reading, Reflection, and Prayer Module 4 – Spanning Tree Protocol Concepts Enabling Activity 4 – Configure a Redundant Network	netacad.com MS Teams SchoolBook		ü	2
Module 5 CLO 3,4 TLO 3,4	Week 8 Mar 10-15	Gospel Reading, Reflection, and Prayer Module 5 - EtherChannel Class Participation 3 - Checkpoint Exam	netacad.com MS Teams SchoolBook		ü	2
Midterm Exam	Week 9 Mar 17-22	Gospel Reading, Reflection, and Prayer Midterm Summative Assessment on Modules 1-5		ü		2
TOTAL				10	8	18

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO5. Explain how to support available and reliable networks using dynamic addressing and first-hop redundancy protocols.	TLO5. Configure a router as DHCPv4 server or client.
CLO6. Configure dynamic address allocation in IPv6 networks.	TLO6. E/plain the operation of SLAAC.
CLO7. Configure WLANs using a WLC and L2 security best practices.	TLO7. Implement port security to mitigate MAC address table attacks
CLO8. Configure switch security to mitigate LAN attacks.	TLO8. Describe WLAN technology and standards

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Onsite / F2F	Asynchronous	Allotted Hours
Module 6 CLO 5,6 TLO 5,6	Week 10 Mar 24-29	Gospel Reading, Reflection, and Prayer **Sustainability Week** Module 6 – DHCPv4 Explain a router as a DHCPv4 server or client Enabling Activity 1 -Implement DHCPv4	netacad.com MS Teams SchoolBook		ü	2
Module 7 CLO 5,6 TLO 5,6	Week 11 Mar 31-Apr 5	Gospel Reading, Reflection, and Prayer Module 7 – SLAAC and DHCPv6 Concepts Explain the operation of SLAAC Enabling Activity 2 – Activity how an IPv6 host can acquire its IPv6 configuration	netacad.com MS Teams SchoolBook	ü		2
Module 7 CLO 5,6 TLO 5,6	Week 12 Apr 7-12	Gospel Reading, Reflection, and Prayer Module 7 – SLAAC and DHCPv6 Concepts Configure a stateful and stateless DHCPv6 server Class Participation 1 – Checkpoint Exam	netacad.com MS Teams SchoolBook	ü		2
Module 8 CLO 5,6 TLO 5,6	Week 13 Apr 14-19	Gospel Reading, Reflection, and Prayer Module 8 – FHRP Concepts Explain the purpose and operation of first hop redundancy protocols Enabling Activity 3 – Learn the config for FHRP	netacad.com MS Teams SchoolBook		ü	2
Module 8 CLO 5,6 TLO 5,6	Week 14 Apr 21-26	Gospel Reading, Reflection, and Prayer Module 8 – FHRP Concepts Explain how HSRP operates Class Participation 2 Checkpoint Exam	netacad.com MS Teams SchoolBook		ü	2
Module 9 CLO 7,8 TLO 7,8	Week 15 Apr 28-May 3	Gospel Reading, Reflection, and Prayer Module 9 – LAN Security Concepts Explain how to use endpoint security to mitigate attacks Enabling Activity 4 – Identify Layer 2 vulnerabilities	netacad.com MS Teams SchoolBook	ü		2
Module 10 CLO 7,8 TLO 7,8	Week 16 May 5-10	Gospel Reading, Reflection, and Prayer Module 10 – Switch Security Configuration Implement port security to mitigate MAC address table attacks Enabling Activity 5 – Switch Security Configuration	netacad.com MS Teams SchoolBook	ü		2
Module 11 CLO 7,8 TLO 7,8	Week 17 May 12-17	Gospel Reading, Reflection, and Prayer Module 11 – WLAN Concepts and Configuration Configure a WLAN to support a remote site Enabling Activity 6 – Troubleshoot common wireless configuration issues Class Participation 3 Checkpoint Exam	netacad.com MS Teams SchoolBook		ü	2

Final Exam	Week 18 May 19-24	Gospel Reading, Reflection, and Prayer Final Term Summative Assessment on Modules 6-11	ü		2
TOTAL			10	8	18

REMINDER: netacad.com PERFORMANCE:

All students taking this course must accomplish the following requirements listed below in order to gain the certificate of completion for Cisco Sem 2: SRWE – Switching Routing, and Wireless Essentials.

- Complete all online chapter exams (Chapter 1-12, PT Practice Skills Assessment) with a minimum grade of 80 for all chapter exams.
- Complete the online final exam with a minimum grade of 90.
- Accomplish the teacher-made test and course feedback online.
- Pass the alternative Hands-on Skills Lab.
- Submit the required PT activities completed to 100%.
- Demonstrate mastery on the interactive activity on key terms and concepts.

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments	50%	Enabling Assessments	50%
Summative Assessment	30%	Summative Assessment	30%
Class Participation	20%	Class Participation	20%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
   a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
   b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
      i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.

ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through the Schoolbook messaging feature.
iii. Email – you may email your professor. (Professor's email will be given during the course orientation)

c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is strictly prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
   a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
   b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
   c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculators during examinations.
   d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

On-Site References

REFERENCES Call number or e-provider	**Reference Material**
9781787121782. 9781787121973.	Harpreet Singh. 2017. Implementing Cisco Networking Solutions : Configure, Implement, and Manage Complex Network Designs. Packt Publishing.
9781800208094. 9781800209985.	Glen D. Singh. 2020. Implementing and Administering Cisco Solutions: 200-301 CCNA Exam Guide : Begin a Successful Career in Networking with CCNA 200-301 Certification. Packt Publishing.
TK 2102.2487 .R762 2017	Cisco Networking Academy companion guide v6. 2017. (Indianapolis, Indiana, USA) : CISCO Press
TK 2101 .C737 2018	Meadows, J. 2018. Communication technology update and fundamentals. (New York ; London) : Routledge Taylor & Francis Group,
IEEE Explore	Panek,C. 2020. Networking Fundamentals. Wiley. Wiley Data and Cybersecurity.
IEEE Explore	Sadiqui, A. 2020. Computer Network Security. Wiley.

Online References

Provider	Reference Material
www.netacad.com	Cisco Networking Academy [accessed: Jan 10, 2025] https://www.netacad.com/portal/resources/course-resources/ccna-srwe
skillsforall.com	Networking Basics [accessed: Jan 10, 2025] https://skillsforall.com/course/networking-basics?userLang=en-US

Videos

VIDEOS Link	Title
https://contenthub.netacad.com/devnet/1.3.1	The Power of Code
https://contenthub.netacad.com/srwe-dl/12.4.1	CAPWAP Operation
https://contenthub.netacad.com/srwe-dl/12.6.1	WLAN Threats
https://contenthub.netacad.com/srwe-dl/12.7.1	Secure WLANs

Prepared: Information Technology Department
2nd Semester 2024-2025

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